

FSHSA36

## Acetic acid, 1N solution in water

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

产品说明:  
Product Description: Acetic acid, 1N solution in water  
Acetic acid, 1N solution in water

Cat No. : SA36-1

Supplier Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Emergency Telephone Number For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

E-mail address begel.sdsdesk@thermofisher.com

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### SECTION 2. HAZARD IDENTIFICATION

Physical State  
Liquid

Appearance  
Colorless

Odor  
No information available

**Emergency Overview**  
Causes severe skin burns and eye damage.

#### Classification of the substance or mixture

|                                   |              |
|-----------------------------------|--------------|
| Skin Corrosion/Irritation         | Category 1 A |
| Serious Eye Damage/Eye Irritation | Category 1   |

#### Label Elements

None required



Signal Word

**Danger**

#### Hazard Statements

H314 - Causes severe skin burns and eye damage

#### Precautionary Statements

Storage

## Acetic acid, 1N solution in water

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

Corrosive. Causes skin and eye burns. Causes serious eye damage.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. . Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component   | CAS No    | Weight % |
|-------------|-----------|----------|
| Water       | 7732-18-5 | 91 - 95  |
| Acetic acid | 64-19-7   | 5 - 9    |

**SECTION 4. FIRST AID MEASURES****General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

**Inhalation**

Remove to fresh air. Get medical attention immediately if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

None reasonably foreseeable. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea

**Self-Protection of the First Aider**

No special precautions required.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous

membranes.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component   | China                                                   | Taiwan                                   | Thailand    | Hong Kong                                                                              |
|-------------|---------------------------------------------------------|------------------------------------------|-------------|----------------------------------------------------------------------------------------|
| Acetic acid | TWA: 10 mg/m <sup>3</sup><br>STEL: 20 mg/m <sup>3</sup> | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup> | TWA: 10 ppm | TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 37 mg/m <sup>3</sup> |

| Component   | ACGIH TLV                   | OSHA PEL                                                                                                 | NIOSH                                                                                                  | The United Kingdom                                                                     | European Union                                                                                                   |
|-------------|-----------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Acetic acid | TWA: 10 ppm<br>STEL: 15 ppm | (Vacated) TWA: 10 ppm<br>(Vacated) TWA: 25 mg/m <sup>3</sup><br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup> | IDLH: 50 ppm<br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup><br>STEL: 15 ppm<br>STEL: 37 mg/m <sup>3</sup> | STEL: 37 mg/m <sup>3</sup><br>STEL: 15 ppm<br>TWA: 10 ppm<br>TWA: 25 mg/m <sup>3</sup> | TWA: 25 mg/m <sup>3</sup> (8h)<br>TWA: 10 ppm (8h)<br>STEL: 50 mg/m <sup>3</sup> (15min)<br>STEL: 20 ppm (15min) |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists  
OSHA - Occupational Safety and Health Administration  
NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Exposure Controls****Engineering Measures**

None under normal use conditions. Ensure that eyewash stations and safety showers are close to the workstation location. .

## Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particle filter

**Small scale/Laboratory use** Maintain adequate ventilation

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                              | Colorless                |                                          |
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Odor</b>                                    | No information available |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | No information available |                                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | No information available |                                          |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | No data available        |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Water Solubility</b>                        | Soluble                  |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Acetic acid                                    | -0.2                     |                                          |

**SAFETY DATA SHEET**

Acetic acid, 1N solution in water

|                           |                          |
|---------------------------|--------------------------|
| Autoignition Temperature  | No data available        |
| Decomposition Temperature | No data available        |
| Viscosity                 | No data available        |
| Explosive Properties      | No information available |
| Oxidizing Properties      | No information available |

**SECTION 10. STABILITY AND REACTIVITY**

|                          |                                                   |
|--------------------------|---------------------------------------------------|
| Stability                | Stable under normal conditions.                   |
| Hazardous Reactions      | None under normal processing.                     |
| Hazardous Polymerization | Hazardous polymerization does not occur.          |
| Conditions to Avoid      | Incompatible products. Excess heat.               |
| Materials to avoid       | Strong bases. Alcohols. Amines. Halogens. Metals. |

**Hazardous Decomposition Products** Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information**

(a) acute toxicity;  
Toxicology data for the components

| Component   | LD50 Oral          | LD50 Dermal | LC50 Inhalation       |
|-------------|--------------------|-------------|-----------------------|
| Water       | -                  | -           | -                     |
| Acetic acid | 3310 mg/kg ( Rat ) | -           | > 40 mg/L ( Rat ) 4 h |

|                                        |                                                           |
|----------------------------------------|-----------------------------------------------------------|
| (b) skin corrosion/irritation;         | No data available                                         |
| (c) serious eye damage/irritation;     | No data available                                         |
| (d) respiratory or skin sensitization; |                                                           |
| Respiratory                            | No data available                                         |
| Skin                                   | No data available                                         |
| (e) germ cell mutagenicity;            | No data available                                         |
| (f) carcinogenicity;                   | No data available                                         |
|                                        | There are no known carcinogenic chemicals in this product |
| (g) reproductive toxicity;             | No data available                                         |
| (h) STOT-single exposure;              | No data available                                         |
| (i) STOT-repeated exposure;            | No data available                                         |
| Target Organs                          | No information available.                                 |

(j) aspiration hazard; Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed** Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea

## SECTION 12. ECOLOGICAL INFORMATION

### Ecotoxicity effects

| Component   | Freshwater Fish                                                                          | Water Flea         | Freshwater Algae | Microtox                                                                                                                                                                        |
|-------------|------------------------------------------------------------------------------------------|--------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acetic acid | Pimephales promelas:<br>LC50 = 88 mg/L/96h<br>Lepomis macrochirus:<br>LC50 = 75 mg/L/96h | EC50 = 95 mg/L/24h | -                | Photobacterium<br>phosphoreum: EC50 =<br>8.8 mg/L/15 min<br>Photobacterium<br>phosphoreum: EC50 =<br>8.8 mg/L/25 min<br>Photobacterium<br>phosphoreum: EC50 =<br>8.8 mg/L/5 min |

### Persistence and Degradability

#### Persistence

Miscible with water, Persistence is unlikely, based on information available, Soluble in water.

### Bioaccumulative Potential

Bioaccumulation is unlikely

| Component   | log Pow | Bioconcentration factor (BCF) |
|-------------|---------|-------------------------------|
| Acetic acid | -0.2    | No data available             |

### Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### Persistent Organic Pollutant

This product does not contain any known or suspected substance

### Ozone Depletion Potential

This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

### Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms.

## SECTION 14. TRANSPORT INFORMATION

### Road and Rail Transport

Not Regulated

### IMDG/IMO

Not regulated

### IATA

Not regulated

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Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component   | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Water       | -                                                   | -                                       | X    | X     | 231-791-2 | X    | X   | X     | X    |      | X    | KE-35400 |
| Acetic acid | X                                                   | X                                       | X    | X     | 200-580-7 | X    | X   | X     | X    | X    | X    | X        |

## National Regulations

## SECTION 16. OTHER INFORMATION

Creation Date 20-May-2010  
Revision Date 15-May-2024  
Revision Summary Not applicable.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

PNEC - Predicted No Effect Concentration

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

**Acetic acid, 1N solution in water**

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Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Physical hazards**

On basis of test data

**Health Hazards**

Calculation method

**Environmental hazards**

Calculation method

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**